# Real-Time Sales Dashboard

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Abstract—In recent years, online classes have been increasingly conducted in various situations. However, in these classes, especially non-face-to-face and large-scale ones, it is more difficult for teachers and students to understand the status of the class during a lecture. To address this issue, we propose a real-time learning analytics dashboard that provides summarized information on teachers' instruction and students' learning activities during lectures. In this article, we introduce the real-time learning analytics dashboard and report its effectiveness through experiments in an online class at our university.

Keywords—real-time analytics, real-time system, online lecture, learning behavior

## I. Introduction

Real-time learning analytics is an important approach for the situational optimization of ongoing lectures. In this approach, educational logs are analyzed to grasp the status of the classroom in real time. Next, according to the learning situation, adaptive feedback is provided to teachers and students. This helpful information allows teachers/students to improve immediately their teaching/learning behaviors [1].

Recently, as the ICT education has become more widespread, online classes have been increasingly conducted in various situations. One online class style involves a teacher conducting a lecture in cyberspace, perhaps using an online meeting system, in the same way as an on-site lecture. This style has advantages such as allowing students to attend from any location and providing students with a more flexible school life. However, in this lecture style, especially in nonface-to-face and large-scale situations, it is more difficult for teachers and students to understand the status of the class compared to on-site lectures. Thus, it is important to enable teachers and students to grasp the status of ongoing lectures [2]. In other words, teachers want to know if students are keeping up with the lecture, and students want to know other students' learning progress and what the teacher is explaining.

Learning analytics dashboards (LADs) provide helpful visual information designed to be understood at a glance to support teachers or students in achieving their intended goals. Depending on the target user, LADs can be classified into three types [7]: those designed for (1) teachers only, (2) students only, and (3) both teachers and students. LADs for teachers provide feedback of their students' learning activities, such as summarized status and at-risk students, to support teachers' instruction [2, 5, 6, 10]. LADs for students present individual learning status and comparisons with the class status to promote self-reflection and improvement [3, 4, 8, 9].

In the field of real-time learning analytics, LADs are also used to display student groups divided by performance in programming courses [6]. In on-site lecture courses, a real-time dashboard can provide students' learning activities from an e-textbook to help teachers to conduct flexible lectures. As a result, students' rate of synchronization with teachers during lectures and highlight usage increased [2]. However, such real-time dashboards mainly focus on realizing immediate and adaptive lecture management by teachers. The effectiveness of direct real-time feedback to students has not yet been verified. In addition, it is also important to investigate what kinds of situation and students are suited to what kinds of dashboard in consideration of the LADs limitations [11].

In this paper, we focus on real-time feedback for students in online classes, where each student joins a lecture remotely using a digital learning environment. We set the following two research questions:

1) Is real-time feedback provided via a LAD effective for students in online lectures? If so, how is it effective?

2) How do students' learning attitudes and perceptions relate to their usage of a real-time LAD?

We focused on non-face-to-face online lectures, where teachers conduct a lesson, and students use an e-book system and listen to the teachers' explanations in cyberspace, such as in an online meeting system. We developed a LAD that allows teachers and students to monitor the learning status of the class during lectures for both online and offline classes. After introducing the early version of our dashboard, we evaluate its effectiveness and discuss the relationship between students' usage and their characteristics through the experimental results from online classes conducted at our university.

## II. System Design and Methods

This section describes our real-time LAD. The dashboard provides several kinds of real-time feedback to students and teachers during lectures based on logs collected via an e-book system called "BookRoll." Both the dashboard and e-book system are a plugin for the university's learning management system called "Moodle" [12] and can be accessed via Moodle (see Fig. 1). The implementation of the real-time LAD relies on BookRoll, so the dashboard reads the log from BookRoll every minute and performs and records aggregation and analysis. Also, when a user accesses the dashboard from Moodle, the user information (ID, role, and course affiliation) is provided so that the dashboard can automatically return an interface and feedback according to the user's attributes.

Thanks to these system linkages, during a lecture, even a nonface-to-face one, the teacher and students can view a textbook on the e-book system and simultaneously open the dashboard to ascertain the progress of the class.

A. BookRoll

BookRoll is an e-book system used to distribute and view digital textbooks. It has standard functions such as browsing, highlights, memos, bookmarking, and a search function. In addition, students can register their understanding by clicking on the "GET IT" or "NOT GET IT" buttons on each page. The operation log of BookRoll, which includes information such as timestamps, user ID, textbook ID, operation type, and page, is recorded online. TABLE 1 shows samples of e-book logs.

B. Real-Time Learning Analytics Dashboard

Our real-time LAD takes the form of a web application so that all teachers and students who attend a lecture course can access it from their own PC through the university's learning management system Moodle. After the teacher or students select the lecture course and textbook, they can open the dashboard, where several kinds of feedback are displayed and updated in real time during the lecture. As feedback, several types of summarized information are provided on the dashboard (see Fig. 2).

1) Class Status

The most basic information is displayed at the top of the dashboard, including the number of students who have the textbook open, the pages most students are viewing, and the page the teacher is explaining. This information allows users to grasp the class activity at a glance.

TABLE 1: SAMPLE OF E-BOOK LOGS

| Timestamp | User_id | E-Book_id | Page | Operation |
|---|---|---|---|---|
| 2020/5/8 10:38:32 | 3243 | A34 | 0 | OPEN |
| 2020/5/8 10:38:34 | 3243 | A34 | 1 | NEXT |
| 2020/5/8 10:38:39 | 3243 | A34 | 2 | NEXT |
| 2020/5/8 10:39:00 | 3243 | A34 | 3 | NEXT |
| 2020/5/8 10:39:27 | 3243 | A34 | 4 | GET IT |

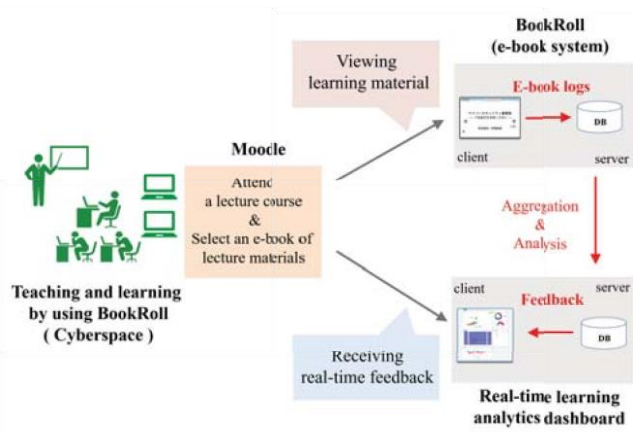



Fig. 1 System overview

is explaining will be displayed, and the user can check the page he or she wants to see, just like an e-book system.

2) Heatmap

The number of people viewing each page at a specific time is visualized as a heat-map chart. The horizontal and vertical axes represent the time and page number of the textbook, respectively, and each cell represents the number of viewing students. This heat-map chart is mainly for the teacher to check the progress of the students.

3) Circular Charts

The upper circular chart shows the percentage of students viewing the same page, the previous page, and the next page based on the teacher's page. The lower circular chart shows the percentage of students that have clicked the “GET IT” and “NOT GET IT” buttons on the teacher’s page. This information enables users to grasp the progress of the class at a glance.

4) Highlights

Highlights drawn by the students on the textbook are aggregated and visualized on the dashboard. The e-book log of highlights contains additional information about the highlight position, so where and how many highlights are drawn on each textbook page can be easily ascertained. To provide an intuitive view of highlight usage, the highlights restored from e-book logs are superimposed on the image of the textbook page, and the color indicates how many highlights have been drawn in that area. If there is no operation on the dashboard, the highlight usage for the page the teacher

5) Remarkable Words

The words that students pay attention to in the textbook are listed according to the degree of attention. This feedback helps users to quickly understand which words in the textbook are important. The e-book logs of highlight and memo operations contain additional information about where the student highlighted and what notes were left. Therefore, the words that drew the students’ attention are defined as words that appeared in the textbook and were highlighted by the students or used in notes left by the students. The degree of attention is defined as the number of times the same word was extracted as the word defined above.

6) Bar Chart

The bar chart shows the understanding of each page of the entire textbook. Students can click the “GET IT” or “NOT GET IT” button as many times as they want before and during the lecture according to their own understanding. Therefore, what is displayed in the understanding charts is the latest understanding evaluation of each student about each page, while the understanding of students who did not press the evaluation button even once is not counted in the charts.

All of this feedback is updated each minute so the teacher and students could grasp the latest status and improve their behavior. For example, if most students are still viewing previous pages, and the understanding evaluation is low, the teacher should slow down and give a supplementary explanation to the students. Additionally, if students lose track of the teacher’s progress in the online lecture, they can refer to the information about the teachers’ page or other students’ highlight usage.

## III. Experiments and Results

### A. Educational Environment

At our university from April 2020, three types of online course were conducted through the university's learning management system: (1) "self-learning style," where a teacher provided lecture materials but not lectures and students studied on their own; (2) "voice-streaming style," where a teacher explained the lecture materials but only his/her voice was streamed, and students engaged in learning while listening to the streaming and viewing the lecture materials with BookRoll, which enabled them to highlight, take notes, and post their understanding status to the teacher; and (3) "video-streaming style," where a video of a teacher explaining the lecture materials was streamed.

### B. Questionnaire about Online Classes

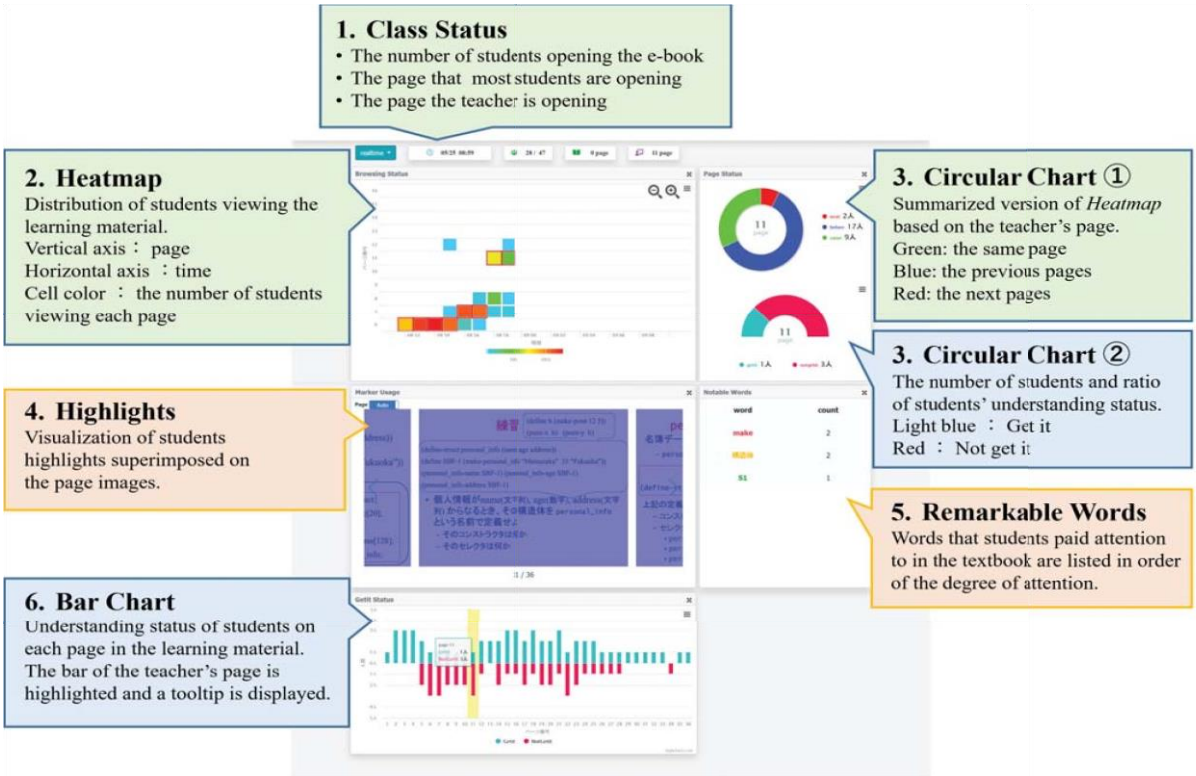
We administered a questionnaire (see Table 2) to 108 undergraduate students of 3 online classes through the learning management system. They were second- or third-year engineering students in the first semester of the 2020 school year. In the questionnaire, participants were asked about their learning attitude, the advantages and disadvantages of online classes, and their satisfaction with online classes compared to on-site classes. The purpose of this questionnaire was to investigate the potential of the proposed dashboard to support their learning during online lectures.

Since the questionnaire consisted of too many questions to present them all in this paper, only the results closely related to the evaluation of the dashboard's potential are shown in Table 3. Each item of questions is evaluated using Likert measurement scale where scores of 1 and 5 represented

Table 2: Questionnaire Overview

| | Purpose | Content | Target | Format |
|---|---|---|---|---|
| Q1 | Status of attending online classes | Anxieties and problems with online classes | | Singlechoice<br>Likert measurement scale<br>Text openended |
| Q2 | Comparison with on-site classes | Learning attitude<br>Advantages and disadvantages (during lectures) | Selflearning | |
| Q3 | | | Voice streaming | |
| Q4 | | | Video streaming | |
| Q5 | Evaluation of online classes | Satisfaction level<br>Important factors in online classes<br>Future university class style | | |

disagreement and agreement, respectively. Compared to onsite lectures about the learning attitude during lectures, students tended to focus on basic learning activities, such as taking notes and listening to teachers' explanations. On the other hand, few students tried to communicate with other students or teachers, which is probably because online classes make such actions more difficult. Regarding the advantages, the item with the highest average was being able to view a lecture while relaxing. This is because online lectures have made it possible for students to view lectures in private spaces such as their homes. Additionally, students found it beneficial to focus more on the teacher's explanation so as not to miss it. Next, students mentioned not knowing the learning status of other students as a disadvantage. However, not being able to talk with other students or having difficulty asking teachers were not considered disadvantages. This is probably because



implementation. The questionnaire showed that most of the students who used the dashboard attached importance to the teacher's explanation, while few students connected the learning situation of other students to their own improvement.

Next, we investigated the learning tendencies of students using the dashboard based on an e-book log analysis. In this paper, we define students utilizing the dashboard's feedback as those who used the dashboard during the lecture in week 8 and gave a score of 4 or more to any questions in TABLE 5. As a criterion for comparing student behavior, we calculated their synchronization rate with teachers, which we defined as the ratio of the number of seconds that the student opened the same page during the teacher's explanation. Fig. 3 shows the distribution of the synchronization rate of three student groups: all students, students utilizing the dashboard, and other students. As shown in Fig. 3, the students utilizing the dashboard maintained a higher synchronization rate in all lectures. In addition, we compared three values of each student: $R_8$ is the student's synchronization rate for week 8, and $R_A$ and $R_M$ are the student's average and max synchronization rates from week 1 to week 7 (see TABLE 6), respectively. In both comparisons, the utilizing group had a higher percentage of students with improved synchronization rates than the other group, possibly because the real-time feedback was used for improvement.

As a case study of students whose learning activities the dashboard positively affected, we visualized two students' learning activities during the lectures from week 6 to week 8. In Fig. 5, each of the six charts represents learning activity based on an e-book by one student during one lecture. The horizontal and vertical axes represent the time and how long the student's page is far from that of the teacher, respectively. The annotation on the charts is the student's e-book operation, and the horizontal lines show the time when the teacher opened the next/previous page.

D. Investigation with Non-negative Matrix Factorization

We applied non-negative matrix factorization (NMF) [8], a technique of component and factor analysis, to elucidate the relationship between students' dashboard evaluations and their learning attitude toward online lectures. To apply NMF,

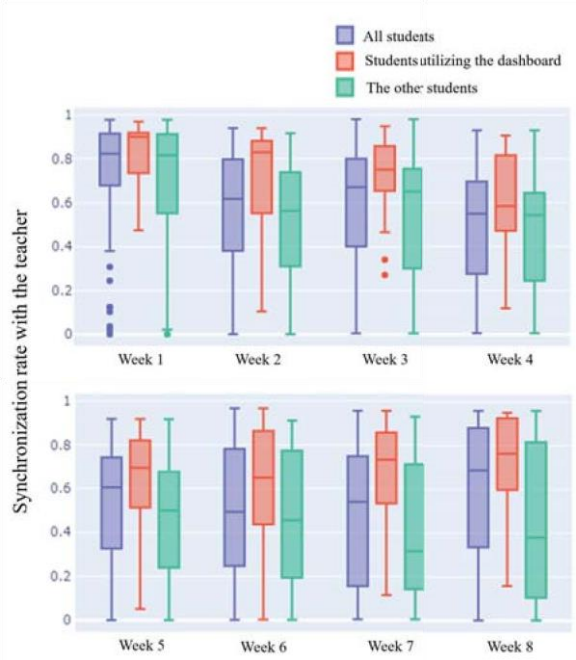



Fig. 3: Synchronization rate with teachers

TABLE 6: IMPROVEMENT OF SYNCHRONIZATION RATE

| | Students Utilizing the Dashboard | Other Students |
|---|---|---|
| Size | 22 | 33 |
| $R_8 > R_A$ | 16 (72%) | 12 (36%) |
| $R_8 > R_M$ | 4 (18%) | 3 (9%) |

we made a "students vs. evaluation" matrix composed of students and their questionnaire answers about online classes and the dashboard evaluation mentioned in the previous sections. Therefore, the elements of the matrix took discrete values from 1 to 5. The NMF decomposed the "students vs. questionnaire" matrix into a "students vs. patterns" matrix and "patterns vs. questionnaire" matrix. These patterns represented latent relationships between the students and the questionnaire answers. In this paper, NMF was performed to find three patterns (see Fig. 4). The number of patterns was a hyper-parameter and was explored from 3 to 5 experimentally.

The features of each pattern are listed in TABLE 7. Through the empirical investigation of student learning attitudes at our university, we interpreted the three patterns as follows. Pattern 1 represented the characteristics of ordinary students who learned by listening to the teacher's explanation in the lecture. Pattern 2 represented that of students in need of support who were strongly anxious about online lectures and gave high evaluations for the real-time feedback. Pattern 3 represented that of students who were highly motivated and actively engaged in self-learning outside the lecture. These results indicated the following two aspects. First, students who felt a strong disadvantage of online lectures tended to value the realtime feedback highly. Second, students who had established their own learning style even in online classes tended not to need the real-time feedback.

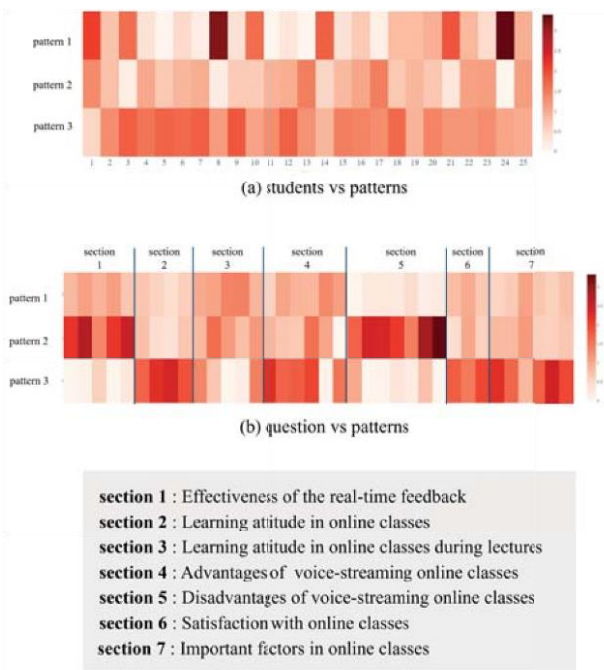



Fig. 4: Decomposed matrix visualized by color scale

TABLE 7: FEATURES OF PATTERNS BY NMF

| | |
|---|---|
| Pattern 1 | Felt weak advantages of online lectures<br>Gave poor evaluation for real-time feedback |
| Pattern 2 | Felt strong disadvantages of online lectures<br>Had low motivation toward online lectures<br>Gave high evaluation of real-time feedback |
| Pattern 3 | Felt strong advantages of online lectures<br>Had high motivation toward online lectures<br>Less interested in other students' learning situations<br>Gave poor evaluation of real-time feedback |

## IV. DISCUSSION AND CONCLUSION

### A. Discussion

In this paper, we set two research questions:

1) Is real-time feedback provided via a LAD effective for students in online lectures? If so, how is it effective?

2) How do students' learning attitudes and perceptions relate to their usage of a real-time LAD?

Regarding the first research question, the experimental results showed that the real-time feedback induced a positive change in some students in terms of their synchronization with the teacher and the use of highlights and the understanding of posting button. However, the improvement depended on the student. Going against our expectation that students would perceive some of the real-time feedback as useful because of the questionnaire about online classes, the dashboard evaluation showed that many students did not find the realtime feedback useful. The following reason could be considered. First, the real-time feedback was descriptive information on the class, so students had to think about when and what to do for improvement, but this is difficult to judge and execute while learning during a lecture. Second, as shown in the text open-ended answer, there was a one-minute lag of the feedback due to implementation limitations, so the students were skeptical of the feedback. This result shows that students want accurate and very quick feedback provided in real time.

Regarding the second research question, the results of the NMF analysis suggested that students who showed a negative reaction to online classes needed real-time feedback, whereas students who showed positive reactions and were able to actively engage in self-learning seemed not to find the real-

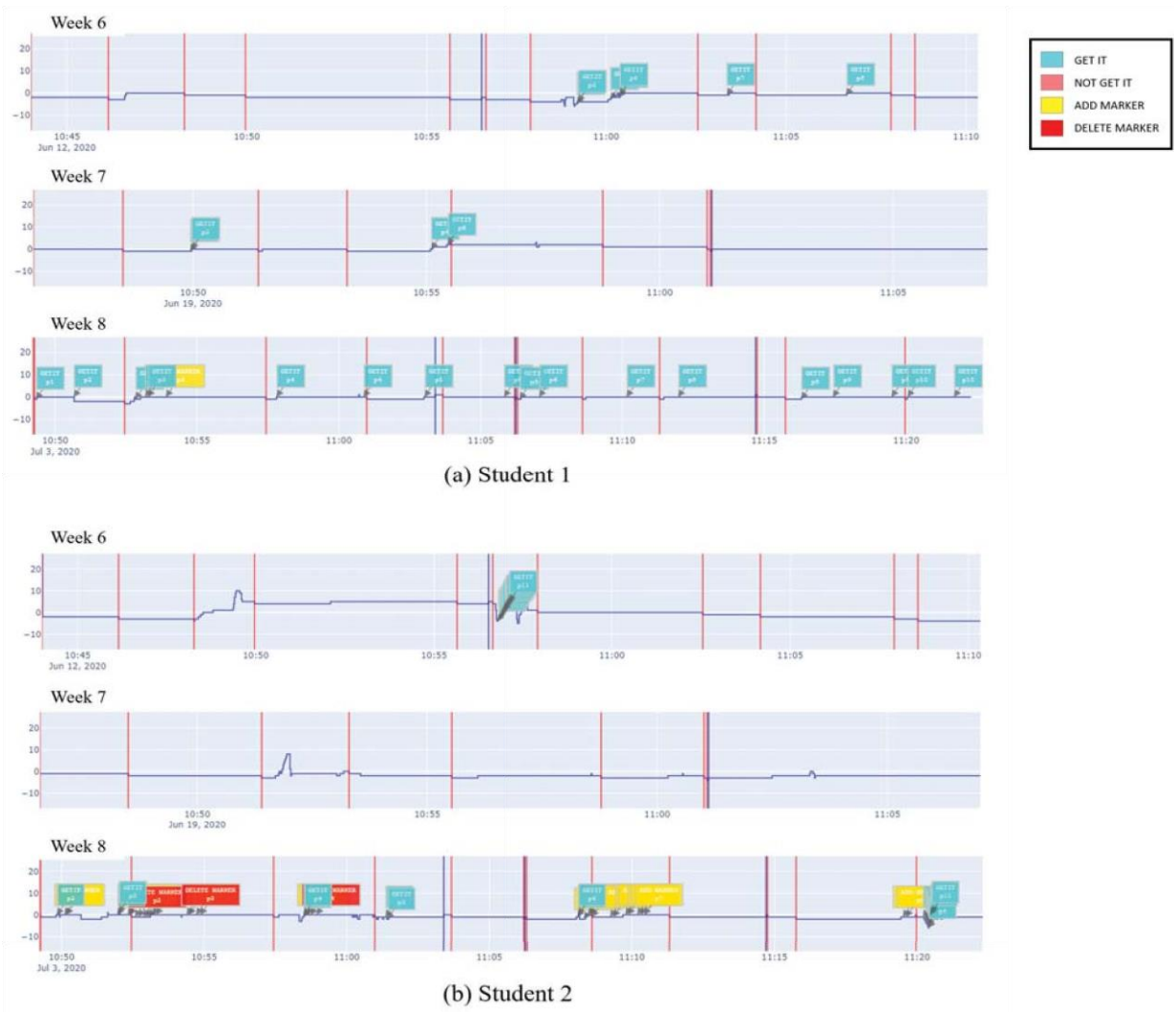



Fig. 5: Visualization of learning activities during a lecture

time feedback useful. This clearly shows that different students need different types of feedback, and this is an issue that needs to be addressed to provide equal support to all students.

## B. Conclusion

In this paper, we introduced an early version of a real-time LAD. In non-face-to-face online classes, it is more difficult for students to grasp the lecture's progress and follow the learning activities than in an on-site class. Therefore, in order for not only teachers but also students to grasp the class activities, our dashboard provides summarized information in real time about the page the teacher is explaining, highlight usage, and students' understanding.

As a result, it was indicated that our system had the following positive effects on synchronization with the teacher and the number of e-book operations, such as highlights and indicating one's understanding during the lecture. Through the cross analytics of the questionnaire about online classes and the dashboard's effectiveness, we found a meaningful relationship between students' learning attitudes and perceptions of online lectures and their real-time feedback evaluation. However, we also found that the presentation of descriptive information was not enough for students to improve their learning during the lecture.

In future works, we will continue to develop and evaluate the effectiveness of our real-time LAD. Specifically, we plan to address system-driven interventions such as real-time alerts and recommendations and individual student feedback. In addition, we will investigate the kind of students and classes in which the proposed system will be more effective. Besides, we are going to compare the effectiveness of our real-time LAD between face-to-face classes and online classes.

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